

How Surfaces Can Represent Functions of Two Variables and Functions of Three Variables

You may have noticed that we have used surfaces to represent functions in two different ways. First, we used a *single* surface to represent a two-variable function $f(x, y)$. Second, we used a *family* of level surfaces to represent a three-variable function $g(x, y, z)$. These level surfaces have equation $g(x, y, z) = c$.

What is the relation between these two uses of surfaces? For example, consider the function

$$f(x, y) = x^2 + y^2 + 3.$$

Define

$$g(x, y, z) = x^2 + y^2 + 3 - z$$

The points on the graph of f satisfy $z = x^2 + y^2 + 3$, so they also satisfy $x^2 + y^2 + 3 - z = 0$. Thus the graph of f is the same as the level surface

$$g(x, y, z) = x^2 + y^2 + 3 - z = 0.$$

In general, we have the following result:

A single surface that is the graph of a two-variable function $f(x, y)$ can be thought of as one member of the family of level surfaces representing the three-variable function

$$g(x, y, z) = f(x, y) - z.$$

The graph of f is the level surface $g = 0$.

Conversely, a single level surface $g(x, y, z) = c$ can be regarded as the graph of a function $f(x, y)$ if it is possible to solve for z . Sometimes the level surface is pieced together from the graphs of two or more two-variable functions. For example, if $g(x, y, z) = x^2 + y^2 + z^2$, then one member of the family of level surfaces is the sphere

$$x^2 + y^2 + z^2 = 1.$$

This equation defines z implicitly as a function of x and y . Solving it gives two functions

$$z = \sqrt{1 - x^2 - y^2} \quad \text{and} \quad z = -\sqrt{1 - x^2 - y^2}.$$

The graph of the first function is the top half of the sphere and the graph of the second function is the bottom half.

Exercises and Problems for Section 12.5

Exercises

1. Match the following functions with the level surfaces in Figure 12.81.

(a) $f(x, y, z) = y^2 + z^2$ (b) $h(x, y, z) = x^2 + z^2$.

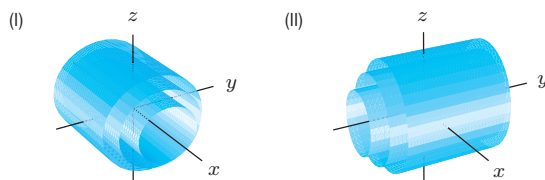


Figure 12.81

2. Match the functions with the level surfaces in Figure 12.82.

(a) $f(x, y, z) = x^2 + y^2 + z^2$
 (b) $g(x, y, z) = x^2 + z^2$.

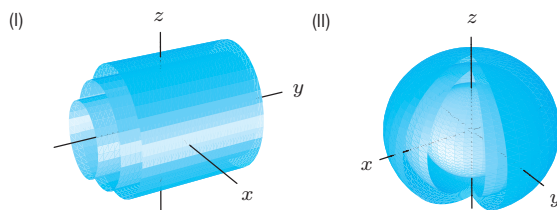


Figure 12.82

3. Write the level surface $x + 2y + 3z = 5$ as the graph of a function $f(x, y)$.
4. Find a formula for a function $f(x, y, z)$ whose level surface $f = 4$ is a sphere of radius 2, centered at the origin.
5. Write the level surface $x^2 + y + \sqrt{z} = 1$ as the graph of a function $f(x, y)$.
6. Find a formula for a function $f(x, y, z)$ whose level surfaces are spheres centered at the point (a, b, c) .
7. Which of the graphs in the catalog of surfaces on page 702 is the graph of a function of x and y ?

Use the catalog on page 702 to identify the surfaces in Exercises 8–11.

$$8. x^2 + y^2 - z = 0 \qquad 9. -x^2 - y^2 + z^2 = 1$$

$$10. x + y = 1 \qquad 11. x^2 + y^2/4 + z^2 = 1$$

In Exercises 12–15, decide if the given level surface can be expressed as the graph of a function, $f(x, y)$.

$$12. z - x^2 - 3y^2 = 0 \qquad 13. 2x + 3y - 5z - 10 = 0$$

$$14. x^2 + y^2 + z^2 - 1 = 0 \qquad 15. z^2 = x^2 + 3y^2$$

Problems

In Problems 16–18, represent the surface whose equation is given as the graph of a two-variable function, $f(x, y)$, and as the level surface of a three-variable function, $g(x, y, z) = c$. There are many possible answers.

16. The plane $4x - y - 2z = 6$
17. The top half of the sphere $x^2 + y^2 + z^2 - 10 = 0$
18. The bottom half of the ellipsoid $x^2 + y^2 + z^2/2 = 1$
19. Suppose the function $f(x, y, z) = 2x - 3y + z - 20$ gives the temperature, in degrees Fahrenheit, at a point (x, y, z) .
 - (a) Describe the isothermal surfaces of f .
 - (b) Calculate and interpret $f_z(0, 0, 0)$.
 - (c) If you are standing at the point $(0, 0, 0)$, in what direction should you move to increase your temperature the fastest?
 - (d) Is $f(x, y) = -2x + 3y + 17$ an isothermal surface of f ? If so, what is the temperature on this isotherm?
20. The balance, B , in dollars, in a bank account depends on the amount deposited, A dollars, the annual interest rate, $r\%$, and the time, t , in months since the deposit, so $B = f(A, r, t)$.
 - (a) Is f an increasing or decreasing function of A ? Of r ? Of t ?
 - (b) Interpret the statement $f(1250, 1, 25) \approx 1276$. Give units.
21. The monthly payments, P dollars, on a mortgage in which A dollars were borrowed at an annual interest rate of $r\%$ for t years is given by $P = f(A, r, t)$. Is f an increasing or decreasing function of A ? Of r ? Of t ?
22. Find a function $f(x, y, z)$ whose level surface $f = 1$ is the graph of the function $g(x, y) = x + 2y$.
23. Find two functions $f(x, y)$ and $g(x, y)$ so that the graphs of both together form the ellipsoid $x^2 + y^2/4 + z^2/9 = 1$.
24. Find a formula for a function $g(x, y, z)$ whose level surfaces are planes parallel to the plane $z = 2x + 3y - 5$.
25. The surface S is the graph of $f(x, y) = \sqrt{1 - x^2 - y^2}$.
 - (a) Explain why S is the upper hemisphere of radius 1, with equator in the xy -plane, centered at the origin.

(b) Find a level surface $g(x, y, z) = c$ representing S .

26. The surface S is the graph of $f(x, y) = \sqrt{1 - y^2}$.
 - (a) Explain why S is the upper half of a circular cylinder of radius 1, centered along the x -axis.
 - (b) Find a level surface $g(x, y, z) = c$ representing S .

27. A cone C , with height 1 and radius 1, has its base in the xz -plane and its vertex on the positive y -axis. Find a function $g(x, y, z)$ such that C is part of the level surface $g(x, y, z) = 0$. [Hint: The graph of $f(x, y) = \sqrt{x^2 + y^2}$ is a cone which opens up and has vertex at the origin.]

28. Describe the level surface $f(x, y, z) = x^2/4 + z^2 = 1$ in words.

29. Describe the level surface $g(x, y, z) = x^2 + y^2/4 + z^2 = 1$ in words. [Hint: Look at cross-sections with constant x , y , and z values.]

30. Describe in words the level surfaces of the function $g(x, y, z) = x + y + z$.

31. Describe in words the level surfaces of $f(x, y, z) = \sin(x + y + z)$.

32. Describe the surface $x^2 + y^2 = (2 + \sin z)^2$. In general, if $f(z) \geq 0$ for all z , describe the surface $x^2 + y^2 = (f(z))^2$.

33. What do the level surfaces of $f(x, y, z) = x^2 - y^2 + z^2$ look like? [Hint: Use cross-sections with y constant instead of cross-sections with z constant.]

34. Describe in words the level surfaces of $g(x, y, z) = e^{-(x^2 + y^2 + z^2)}$.

35. Sketch and label level surfaces of $h(x, y, z) = e^{z-y}$ for $h = 1, e, e^2$.

36. Sketch and label level surfaces of $f(x, y, z) = 4 - x^2 - y^2 - z^2$ for $f = 0, 1, 2$.

37. Sketch and label level surfaces of $g(x, y, z) = 1 - x^2 - y^2$ for $g = 0, -1, -2$.

Strengthen Your Understanding

In Problems 38–40, explain what is wrong with the statement.

38. The graph of a function $f(x, y, z)$ is a surface in 3-space.
39. The level surfaces of $f(x, y, z) = x^2 - y^2$ are all saddle-shaped.
40. The level surfaces of $f(x, y, z) = x^2 + y^2$ are paraboloids.

In Problems 41–44, give an example of:

41. A function $f(x, y, z)$ whose level surfaces are equally spaced planes perpendicular to the yz -plane.
42. A function $f(x, y, z)$ whose level sets are concentric cylinders centered on the y -axis.
43. A nonlinear function $f(x, y, z)$ whose level sets are parallel planes.
44. A function $f(x, y, z)$ whose level sets are paraboloids.

Are the statements in Problems 45–54 true or false? Give reasons for your answer.

45. The graph of the function $f(x, y) = x^2 + y^2$ is the same as the level surface $g(x, y, z) = x^2 + y^2 - z = 0$.

46. The graph of $f(x, y) = \sqrt{1 - x^2 - y^2}$ is the same as the level surface $g(x, y, z) = x^2 + y^2 + z^2 = 1$.
47. Any surface which is the graph of a two-variable function $f(x, y)$ can also be represented as the level surface of a three-variable function $g(x, y, z)$.
48. Any surface which is the level surface of a three-variable function $g(x, y, z)$ can also be represented as the graph of a two-variable function $f(x, y)$.
49. The level surfaces of the function $g(x, y, z) = x + 2y + z$ are parallel planes.
50. The level surfaces of $g(x, y, z) = x^2 + y + z^2$ are cylinders with axis along the y -axis.
51. A level surface of a function $g(x, y, z)$ cannot be a single point.
52. If $g(x, y, z) = ax + by + cz + d$, where a, b, c, d are nonzero constants, then the level surfaces of g are planes.
53. If the level surfaces of g are planes, then $g(x, y, z) = ax + by + cz + d$, where a, b, c, d are constants.
54. If the level surfaces $g(x, y, z) = k_1$ and $g(x, y, z) = k_2$ are the same surface, then $k_1 = k_2$.

12.6 LIMITS AND CONTINUITY

The sheer face of Half Dome, in Yosemite National Park in California, was caused by glacial activity during the Ice Age. (See Figure 12.83.) As we scale the rock from the west, the height of the terrain rises abruptly by nearly 5000 feet from the valley floor, 2000 feet of it vertical.

If we consider the function h giving the height of the terrain above sea level in terms of longitude and latitude, then h has a *discontinuity* along the path at the base of the cliff of Half Dome. Looking at the contour map of the region in Figure 12.84, we see that in most places a small change in position results in a small change in height, except near the cliff. There, no matter how small a step we take, we get a large change in height. (You can see how crowded the contours get near the cliff; some end abruptly along the discontinuity.)

This geological feature illustrates the ideas of continuity and discontinuity. Roughly speaking, a function is said to be *continuous* at a point if its values at places near the point are close to the value at the point. If this is not the case, the function is said to be *discontinuous*.

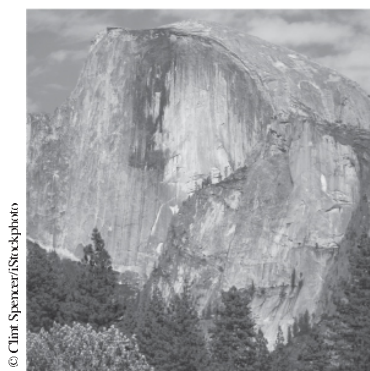


Figure 12.83: Half Dome in Yosemite National Park

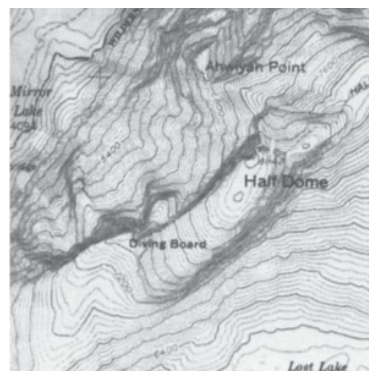


Figure 12.84: A contour map of Half Dome